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Digital Signal Processing
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Re: Submission of manuscript – "Fisher–Rao-Guided Generative Steganography: Geodesic Embedding and Detection Analysis"

Dear Editor,

We are pleased to submit the manuscript entitled "Fisher–Rao-Guided Generative Steganography: Geodesic Embedding and Detection Analysis" for consideration in *Digital Signal Processing*.

The manuscript treats generative and coverless steganography as a statistical signal-generation and detection problem. Its central question is whether a message-bearing generator should be analyzed only through the endpoint divergence between cover and stego distributions, or whether the embedding trajectory through the generator parameter space also matters for accumulated detectability. We show that the trajectory does matter: in the small-step regime, the sequential Kullback–Leibler cost of an embedding path is governed by Fisher path energy, and Fisher–Rao geodesics provide locally optimal embedding trajectories.

We believe the paper fits the scope of *Digital Signal Processing* because it contributes to statistical signal processing, detection theory, information security, and mathematical methods with direct application to signal-processing models. Fisher–Rao geometry is not presented as an end in itself; it is used as the intrinsic sensitivity metric for controlling and analyzing low-detectability generative steganographic embedding.

The main contributions are:

- a detection-aware model of generative steganographic embedding as controlled motion on a statistical signal manifold;
- a sequential-KL path-cost formulation showing why endpoint divergence alone can miss accumulated embedding detectability;
- a geodesic embedding result proving local optimality of Fisher–Rao paths in the small-step regime;
- a Natural-Gradient Steganography algorithm with convergence analysis under geodesic smoothness and strong-convexity assumptions;
- curvature-based robustness diagnostics for embedding-path sensitivity;
- a Neyman–Pearson detection interpretation connecting the steganalyzer's statistic to the cover-to-stego displacement in exponential-family signal models.

The manuscript is methodological and theoretical. It is not submitted as a complete operational image-steganography pipeline; rather, it provides a signal-processing framework for designing and analyzing embedding paths in generative statistical models. Numerical experiments on Gaussian and Gaussian-mixture signal models illustrate the predicted KL reduction, natural-gradient speedups, and curvature-driven sensitivity effects.

The manuscript has not been published elsewhere and is not under consideration by any other journal. All authors have approved this submission. The authors declare no competing interests. AI-assisted language tools were used to improve readability; all scientific content was reviewed and edited by the authors, who take full responsibility for the manuscript.

Respectfully yours,

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on behalf of all co-authors: P.R. Mama Assandje, S.A. Ebélé, S.W. Mossebo Tcheunteu, A.U. Ewane, D.B.M. Fotso, R. Ndoundam.